

SYNTHESIS OF POTASSIUM TRISOXALATOFERRATE(III) TRIHYDRATE

PURPOSE

The purpose of this experiment is to synthesize a transition metal coordination compound.

BACKGROUND

Read pages 1016 – 1033 in your text to get a fuller understanding of coordination compounds than the brief basics given below.

Coordination compounds are formed when (1) a neutral metal atom, acting as a Lewis acid, reacts with some neutral molecules, acting as Lewis bases; or when (2) a metallic cation, acting as a Lewis acid, reacts with any of a variety of organic or inorganic molecules, cations, or anions, acting as Lewis bases. These Lewis bases are called *ligands*. (Lewis acids are electron pair acceptors and Lewis bases are electron pair donors. You will have to draw the Lewis structure of a molecule or ion to determine if it is capable of acting as a Lewis base or ligand.) Some examples of ions or molecules that typically act as ligands or Lewis bases are:

Neutral ligands: H_2O , NH_3 , ethylenediamine ($\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$), CO , NO , $\text{C}_5\text{H}_5\text{N}$ (pyridine), $\text{P}(\text{OCH}_2)_3\text{CH}_3$

Anionic ligands: Cl^- , Br^- , I^- , OH^- , CN^- , SCN^- , SO_4^{2-} , CO_3^{2-} , $\text{C}_2\text{O}_4^{2-}$, EDTA^{4-}

Cationic ligands: NO^+ , NH_2NH_3^+

Examples of Case 1 above (when a neutral metal atom reacts with some neutral molecules) are neutral coordination compounds like:

$\text{Ni}(\text{CO})_4$ Tetracarbonylnickel(0) $\text{Fe}(\text{CO})_5$ Pentacarbonyliron(0) $\text{Cr}(\text{CO})_6$ Hexacarbonylchromium(0)

$\text{Ni}(\text{PF}_3)_4$ Tetrakis(trifluorophosphine)nickel(0) $\text{C}_7\text{H}_8\text{Mo}(\text{CO})_3$ Tricarbonylcycloheptatrienemolybdenum(0)

Examples of Case 2 above (when a metallic cation reacts with neutral molecules, anions, or, rarely, cations) are more numerous:

$\text{K}_3[\text{Fe}(\text{CN})_6]$ Potassium hexacyanoferrate(III)

$[\text{Fe}(\text{NH}_3)_6]\text{Cl}_2$ Hexaammineiron(II) chloride

$[\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{Pt}(\text{NH}_3)\text{Cl}_3]$ Chlorotriammineplatinum(II) trichloroammineplatinate(II)

Sometimes the coordination compound contains a simple cation and a complex anion, as in the first case above. Sometimes the cation is complex and the anion is simple, as in example two, and, finally, example three above illustrates a coordination compound where both the cation and the anion are complex ions.

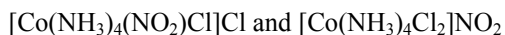
Some ligands, when they bond to a transition metal cation, can donate more than one pair of electrons to the metal if the geometry of the ligand and the spacing of available electron pairs on the ligand are appropriate. Such ligands are *polydentate* (many-toothed) ligands, and they must be able to wrap themselves around the central metal ion in order to bring more than one electron pair into proper orientation toward the metal. The EDTA^{4-} ion (ethylenediaminetetraacetate ion, sometimes carelessly called the “edetate” ion) is a hexadentate ligand. The ligand you are using in this synthesis, the oxalate ion ($\text{C}_2\text{O}_4^{2-}$ ion), is a bidentate ligand.

For an understanding of the theories of the nature of the bonding in coordination compounds and complex ions, consult pages 1024 – 1031 in your text. For a discussion of the magnetic properties and colors of coordination compounds or complex ions, consult pages 1028 – 1030 in your text.

Isomerism commonly occurs in coordination compounds (see pages 1021 – 1024 in your text). There are many different types of isomerism:

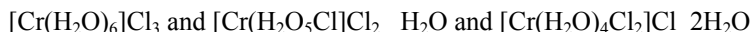
1. Ionization Isomerism

In this case, an interchange of the anion and a negative ligand in a complex ion occurs, *e.g.*,



2. Hydrate Isomerism

In this case, an interchange of an anion with a neutral ligand (H_2O) in a complex ion occurs:

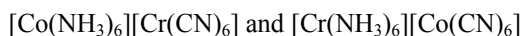


3. Ligand Isomerism

In ligand isomerism, the isomerism arises from isomerism within a ligand. In $[\text{Co}(\text{en})_2(\text{CH}_3\text{C}_6\text{H}_4\text{NH}_2)\text{Cl}]\text{Cl}_2$, for example, the $\text{CH}_3\text{C}_6\text{H}_4\text{NH}_2$ ligand can exist as either the *o*-toluidine (or 2-aminotoluene), *m*-toluidine (or 3-aminotoluene), or *p*-toluidine (4-aminotoluene) ligand.

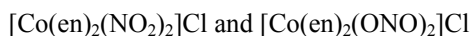
4. Coordination Isomerism

If a coordination compound is composed of both a complex cation and a complex anion, isomers may result from the interchange of ligands between the coordination centers:



5. Linkage Isomerism

In linkage isomerism, the same ligand may attach to the metal coordination center using either of two different atoms:



In the first example, the nitro ligand coordinates to the cobalt(III) ion through the N atom, but in the second example, the coordination of the nitro ligand is through the O atom.

6. Stereoisomerism

In coordination compounds in which the coordination number of the central atom is four or more, *geometrical* (or *cis/trans* isomerism) or *optical isomerism* may occur. In geometrical or *cis/trans* isomerism, two ligands of the same type (e.g., NH_3) may coordinate to the central metal ion either across (*trans*) from each other or adjacent (*cis*) to each other. Optical isomerism requires that a compound exist in both a “left-handed” or “right-handed” form, *i.e.*, that there exist two nonsuperimposable mirror images or *enantiomers*. One enantiomer rotates the plane of polarized light to the left and the other enantiomer rotates the plane of polarized light to the right. Consider the $[\text{Co}(\text{en})_2\text{Cl}_2]^{3+}$ ion. Convince yourself that this ion (1) has two geometrical (*cis/trans*) isomers and that (2) the *cis* isomer exhibits optical isomerism but the *trans* isomer does not.

To write the formula of a coordination compound, three rules must be followed (see page 1018 in text). Two are the same as for writing the formula of a simple ionic compound: (1) place the cation before the anion; and (2) balance the charge of the cation(s) and anion(s). The third rule applies to writing the formula of a complex ion: write the symbol of the metal, followed in order by the cationic ligands, neutral ligands and, finally, anionic ligands. Place the formula of the entire complex ion with brackets.

The following rules must be followed to name coordination compounds (see page 1019 in text and note that *rule 2, as stated in the text, is incorrect and does not follow IUPAC recommendations*):

1. Name the cation before the anion.
2. Within the complex ion, name the ligands in the following order: anionic, neutral, cationic. Then name the metal ion.
3. The names of neutral ligands usually bear the name of the lone molecule, but there are exceptions. For example, when naming H_2O , NH_3 , CO , and NO as ligands, the names become *aquo*, *ammine*, *carbonyl*, and *nitrosyl*.
4. To obtain the name of an anionic ligand, drop “-ide” from the root name and add “-o.” For example, F^- is named *fluoro* instead of *fluoride*.
5. If there are several anionic ligands, name them in the order: H^- , O^{2-} , OH^- , other simple ions, polyatomic ions.
6. If there are several neutral ligands, name them in the order: H_2O , NH_3 , other inorganic ligands, organic ligands in alphabetical order.
7. Use a numerical prefix to designate the number of a particular ligand (but disregard the prefix when determining alphabetical order): *di-*, *tri-*, *tetra-*, *penta-*, *hexa-*, etc. for 2, 3, 4, 5, and 6, respectively. However, if the ligand

- name is complicated, the prefixes *bis-*, *tris-*, *tetrakis-* are preferred for 2, 3, and 4, respectively.
- The oxidation state of the central metal ion is given by a Roman numeral in parentheses (Stock system) if the central metal ion can exhibit more than one oxidation state.
 - In a complex anion, the ending of the name of the metal is dropped and the suffix “-ate” is added.

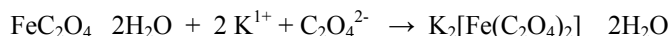
EXPERIMENTAL PROCEDURE

1. Preparation of Iron(II) Oxalate (FeC₂O₄)

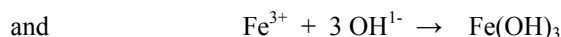
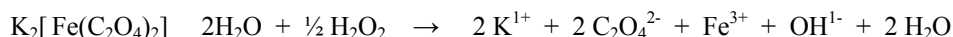
- In a 50-mL beaker, warm (don't heat to boiling!) 9 mL of distilled water, and add 3 drops of dilute (3M) H₂SO₄.
- Dissolve 2.5 g of ferrous ammonium sulfate hexahydrate in the acidified warm water prepared in step (a).
- With stirring, add 12.5 mL of an oxalic acid (H₂C₂O₄) solution containing 10 g of oxalic acid per 100 mL of solution to the solution of ferrous ammonium sulfate prepared in step (b).
- Cautiously heat the mixture to boiling, stirring carefully to prevent bumping.
- Withdraw the heat, and allow the granular yellow precipitate of iron(II) oxalate dihydrate to settle.
- Decant the clear supernatant liquid.
- Add 9 mL of hot distilled water to the iron(II) oxalate dihydrate, stir, allow the solid to settle, and decant the supernate as before.
- Wash the iron(II) oxalate onto a suction filter with hot distilled water, and wash the yellow solid with a few mL of acetone.

2. Preparation of Potassium Trisoxalatoferrate(III) Trihydrate, K₃[Fe(C₂O₄)₃] · 3 H₂O

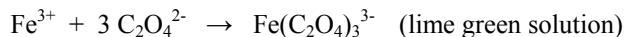
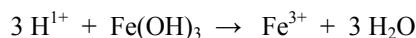
- Weigh approximately 0.4 g of dried iron(II) oxalate dihydrate prepared in step (1) above into a 100-mL beaker, and wet the solid with a few drops of distilled water.
- In a 50-mL beaker, dissolve 0.90 g of potassium oxalate hydrate [K₂C₂O₄ · H₂O] in 4 mL of warm distilled water and pour this solution into the solution in step (a). At this point, an orange intermediate iron(II) complex is formed:



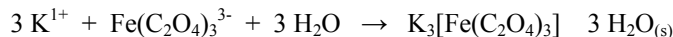
- Warm the solution from step (b) to about 40°C, then remove from the heat and add 5.0 mL of 3% hydrogen peroxide (H₂O₂) **dropwise while stirring continuously**. The hydrogen peroxide oxidizes iron (II) to iron(III), and a brown precipitate of Fe(OH)₃ is formed.



- Heat the mixture from step (c) to boiling, and add 3.0 mL of 1.0 M oxalic acid (H₂C₂O₄) solution. A clear lime-green solution should appear. If it does not, add a few more drops of 1.0 M oxalic acid. Upon adding the oxalic acid, the iron(III) hydroxide dissolves, and the trisoxalatoferrate(III) ion [Fe(C₂O₄)₃]³⁻ forms:



- Cool the solution from step (d) to approximately 50°C, and add 5 mL of 95% ethanol (C₂H₅OH). Cover, and cool in an ice/water bath. Cool for 15 minutes *after* reaching 5°C. The product, potassium trisoxalatoferrate(III) is very soluble in water but only slightly soluble in ethanol.



- Transfer the product to a Buchner funnel attached to a suction flask, and wash the product twice with separate 5-mL portions of 50/50 ethanol/water. Discard the filtrate in the hazardous waste container. Leave the product on a watch glass to dry in your locker. The product is photosensitive. At the beginning of the next lab, weigh your product, and calculate the percent yield.

POSTLAB QUESTIONS

1. Write an overall balanced equation for the synthesis of potassium trisoxalatoferrate(III) trihydrate starting with iron(II) oxalate dihydrate.
2. State the actual yield, and calculate the theoretical yield and the percent yield of your potassium trisoxalatoferrate(III) trihydrate? SHOW CALCULATIONS.
3. What is the name for the orange intermediate $\text{K}_2[\text{Fe}(\text{C}_2\text{O}_4)_2] \cdot 2\text{H}_2\text{O}$?
4. Is the oxalate ion, $\text{C}_2\text{O}_4^{2-}$, a monodentate or bidentate ligand in trisoxalatoferrate(III) trihydrate?
5. What is the geometry of the trisoxalatoferrate(III) ion? Predict whether or not this ion will exhibit optical isomerism.
6. Provide systematic IUPAC names for each of the following coordination compounds:
 - a. $\text{K}_4[\text{Fe}(\text{CN})_6]$
 - b. $[\text{Co}(\text{NH}_3)_4\text{SO}_4](\text{NO}_3)$
 - c. $[\text{Cr}(\text{NH}_3)_4(\text{H}_2\text{O})_2][\text{Co}(\text{CN})_6]$
 - d. $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 - e. $(\text{NH}_4)_3[\text{ZrF}_7]$
 - f. $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})(\text{Cl})]\text{Br}_2$

